

Test Matrix — Problem 6: Intrusion Detection

CCP-CNS2026 · Week 4 · Team: Muhammad Ahmed, Hamna Rizwan, Eshal Tanveer · 02 June 2025

Total Tests: 14 **PASS: 14** **FAIL: 0** **PENDING: 0** Pass Rate: 100%

ID	Test Name	Procedure	Expected Result	Actual Result	Status
A — File Integrity & Baseline Verification					
A-01	PCAP file integrity check <i>File verification</i>	Open Wireshark → Statistics → Capture File Properties → record SHA-256 hash	Hash matches: 33e274d7246f4eca8fbe7337c465c7ded077cd650d5d4d42481cad822c962741	Hash confirmed matching	PASS
A-02	Baseline packet count <i>File verification</i>	Statistics → Capture File Properties → verify total packet count	Total packets = 48,877 over 34 min 32 sec	48,877 packets confirmed	PASS
A-03	Capture time range <i>File verification</i>	Check first and last packet timestamps in File Properties	First: 2025-06-13 20:33:55 — Last: 2025-06-13 21:08:28	Timestamps confirmed correct	PASS
B — Attacker & Victim Identification					
B-01	Top talker identification <i>Conversations analysis</i>	Statistics → Conversations → IPv4 tab → sort by Packets descending	Top conversation: 10.6.13.133 ↔ 83.137.149.15 with 35,485 packets / 34 MB	35,485 packets / 34 MB confirmed as top entry	PASS
B-02	Victim IP verification <i>IP identification</i>	Apply filter: ip.dst == 10.6.13.133 — confirm packet count and inbound traffic	30,219 packets displayed (61.8%) — all destined for victim	30,219 packets shown — filter confirmed	PASS
B-03	Attacker IP verification <i>IP identification</i>	Apply filter: ip.src == 83.137.149.15 — confirm attacker traffic volume	23,281 packets displayed (47.6%) — all from attacker IP	23,281 packets confirmed	PASS
C — Attack Classification Evidence					
C-01	Port 443 dominance test <i>Protocol analysis</i>	Apply filter: tcp.port == 443 — record displayed	41,941 packets (85.8% of all traffic) on port 443	41,941 packets / 85.8% confirmed	PASS

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		packet count and percentage			
C-02	SYN packet count test <i>TCP flag analysis</i>	Apply filter: tcp.flags.syn == 1 — count total SYN/SYN-ACK packets	426 SYN packets (0.9%) — low ratio confirms sustained session not SYN flood	426 packets confirmed — rules out pure SYN flood	PASS
C-03	TLS handshake verification <i>Protocol analysis</i>	Filter: ip.src == 83.137.149.15 → locate packet 8567 → expand TCP and TLS layers	Packet 8567 is SYN,ACK on port 443 at 157.661094s; followed by TLSv1.2 handshake (Client Hello SNI=windows.php.net, Server Hello, Certificate)	Full TLS 1.2 handshake captured and confirmed	PASS
C-04	Traffic asymmetry ratio <i>Volume analysis</i>	In Conversations view, note Bytes A→B vs Bytes B→A for attacker-victim pair	Attacker → Victim: 33 MB vs Victim → Attacker: 754 KB (44:1 ratio)	33 MB vs 754 KB ratio confirmed — strong asymmetry	PASS
C-05	DNS camouflage check <i>DNS analysis</i>	Apply filter: dns — search for query resolving to 83.137.149.15 near timestamp 157s	Packet 8565 at 157.513273s — DNS response resolves windows.php.net → 83.137.149.15	DNS camouflage via windows.php.net confirmed	PASS
C-06	Single-session dominance check <i>Attack classification</i>	Confirm no other IP pair exceeds 10% of total traffic in conversations view	Second highest conversation has only 1,727 packets vs 35,485 — single-session attack confirmed, not distributed	No other session exceeds 4% — DDoS ruled out	PASS
D — First Suspicious Packet & Timeline					
D-01	First packet timestamp test <i>Timeline forensics</i>	Filter ip.src == 83.137.149.15 → go to first result — record packet number and timestamp	Packet #8567 at 157.661094s — SYN,ACK from port 443 to victim port 52527	Packet 8567 at 157.661094s confirmed	PASS
D-02	Attack duration measurement <i>Timeline forensics</i>	In conversations view, note Rel. Start and Duration for top conversation (stream 58)	Rel. Start: 157.513925s, Duration: 136.0165s — attack ran ~136 seconds	136s duration confirmed from conversations view	PASS
E — Mitigation Validation (Logical / Firewall Rule Tests)					
E-01	Firewall rule scope test <i>Mitigation logic</i>	Verify that blocking 10.6.13.133 → 83.137.149.15 on	Rule is host-specific (single src+dst+port) — no legitimate traffic from other hosts is blocked	Rule confirmed host-specific —	PASS

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		TCP/443 would eliminate 85.8% of malicious traffic without affecting other hosts		no collateral impact	
E-02	DNS sinkhole effectiveness test <i>Mitigation logic</i>	Confirm windows.php.net is not a legitimate Microsoft domain — verify it resolves to attacker IP only	microsoft.com and windows.com are legitimate; windows.php.net is attacker-controlled — safe to sinkhole	Domain confirmed attacker-controlled — sinkhole safe	PASS
E-03	IDS rule trigger test <i>Mitigation logic</i>	Write Snort/Suricata-style rule: alert on single-session outbound >5 MB on port 443 within 10 minutes from internal host	Threshold (5 MB / 10 min) would be triggered by attacker session (33 MB / 136 sec) within first ~20 seconds	Rule would trigger at ~20s into attack — early detection confirmed	

All tests performed on instructor-provided PCAP: 2025-06-13-traffic-analysis-exercise.pcap (42 MB, SHA-256 verified)